Sperm Motility

This **Fiji/ImageJ plugin** is intended to provide an automated analysis of sperm motility, by tracking the live fluorescent sperm over time. This plugin takes as input a folder of TIF images, and outputs a CSV file with raw tracking results. It can be used to track any small round fluorescent dots in time-lapse images, as it is simply an automated version of the Trackmate plugin.

User guide

1. Installation

a. Fiji download

First of all, you need to have **Fiji** installed on your computer. You can download it here.

> **Note:** When extracting the downloaded .zip file, it is best to choose the root of your drive as the location (e.g. `C:/Fiji.app` on Windows or `/Applications/Fiji.app` on macOS).

b. Dependencies

Next, you will need to add the `csv-commons.jar` dependency to the **jars** folder of your Fiji installation. To find this folder, first find the Fiji file location by right-clicking on the app and selecting *open file location*. Then drag the `.jar` file into the **jars** folder. This file is required for csv file handling.

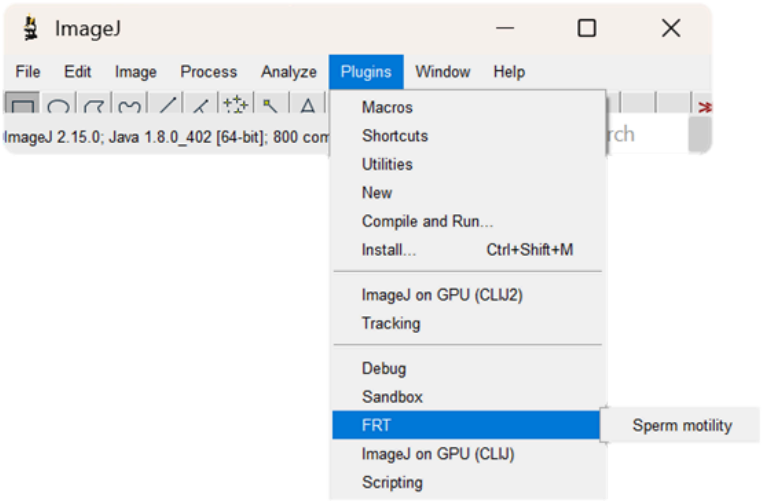
The only other required dependency is `clij2`, which is already included in the Fiji installation. To check if it is installed, open Fiji and go to the *Help* menu and select *Update...*. Then click *Manage update sites* and make sure that both `clij` and `clij2` update site are checked. Click on *Close*, then click on *Apply Changes*, and restart Fiji to apply the changes. Follow this link for more guidance on the clij2 installation.

c. Plugin download

The plugin is available for download here. Feel free to rename the file to something like *SpermMotility.jar* for easier identification.

Once downloaded, you can drag the `.jar` file into a specific folder of the Fiji installation. First, find the Fiji file location by right-clicking on the app and selecting *open file location*. Then drag the `.jar` file into the **plugins** folder. The complete path should be something like this: `C:\Fiji.app\plugins`.

Restart Fiji if it was open and the plugin will be available in the *Plugins* menu:



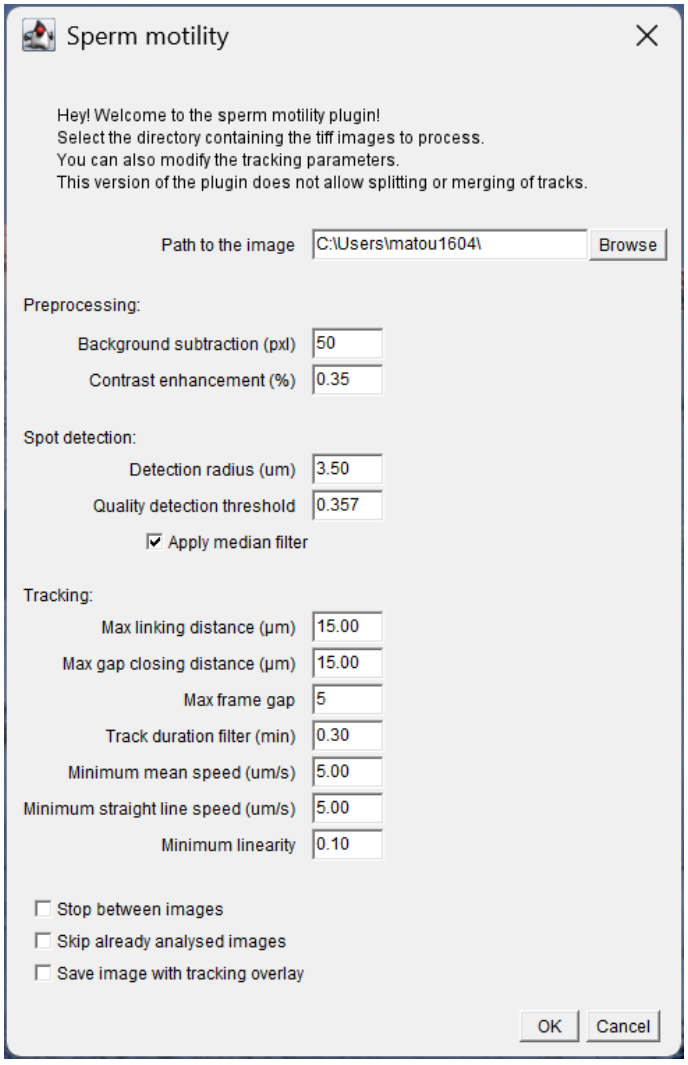
2. Running the plugin

When running the plugin, a user interface window will appear.

Select the folder containing the TIFF images you want to analyze. A results folder will be created in the same directory as the input folder, and the results will be saved there.

You can then modify the parameters of the preprocessing, spot detection, and tracking of the images:

- **Background subtraction radius**: the radius of the background subtraction circle, in pixels.
- **Contrast enhancement**: saturation accepted for contrast enhancement, meaning, the percentage of pixels that are clipped at the very dark and very bright ends of the histogram. Only visual/for display purposes, does not change the pixel values.
- **Detection radius**: the radius of the detection circle, in µm.
- **Quality detection threshold**: the minimum quality of the spot detection. This value is subjective to each experiment, and should be adjusted for each dataset in order to correctly detect the sperm. A higher value will result in fewer detections, while a lower value will result in more detections.
- **Apply median filter**: if checked, a median filter will be applied to the image before detection. It can



help to reduce noise and improve the detection of the sperm.

- **Max linking distance**: the maximum distance between two detections to be considered as the same sperm, in µm.
- **Max gap closing distance**: the maximum distance between two detections to be considered as the same sperm, if they are not detected in consecutive frames, in µm.
- **Max frame gap**: the maximum number of frames between two detections to be considered as the same sperm, if they are not detected in consecutive frames.
- **Track duration filter**: the minimum duration of a track to be considered as valid, in minutes.
- **Minimum mean speed**: the minimum mean speed of a track to be considered as motile, in µm/s.
- **Stop between images**: if checked, the plugin will stop after each image is analyzed, and you will have to click *OK* to continue to the next image. This can be useful for checking the results of the analysis, and for adjusting the parameters if needed.
- **Skip already analyzed images**: if checked, the plugin will skip the images that have already been analyzed. This can be useful for adjusting the parameters and reanalyzing the images without having to delete the results folder.
- **Save image with tracking overlay**: if checked, the plugin will save a screenshot of the image being analyzed, with the tracks overlaid on top of the image. The screenshot will be saved in the results folder, with the same name as the input image, but with a `.png` extension.

During the analysis, the plugin will display the image being analysed. If you want to see the results of the analysis, you can check the ***Stop between images*** box.

Understanding the plugin

1. Trackmate

This plugin is based on the Trackmate plugin, which is a powerful tool for tracking objects in images. It is a plugin written in Java that scripts the Trackmate plugin to automate it. It uses the CLIJ2 library for GPU-accelerated image processing, which allows for fast processing of large images.

2. Results

The results of the analysis are saved in a CSV file in the results folder. The file `tracks_nameofinpoutimages.csv` contains the following columns:

- [x] LABEL: the label of the track.
- [x] TRACK_ID: the ID of the track.
- [x] NUMBER_SPOTS: the number of spots in the track.
- [x] NUMBER_GAPS: the number of frame gaps in the track.
- [x] TRACK_DURATION: the duration of the track in minutes.
- [x] TRACK_DISPLACEMENT: the displacement of the track in µm.
- [x] TRACK_MEAN_SPEED: the mean speed of the track in µm/s.
- [x] TOTAL_DISTANCE_TRAVELED: the total distance traveled by the spot in µm.
- [x] MEAN_STRAIGHT_LINE_SPEED: the mean straight line speed of the track in µm/s.

- [x] LINEARITY_OF_FORWARD_PROGRESSION: the linearity of forward progression of the track, calculated as the ratio between the displacement and the total distance traveled.
- [x] MOTILE: a boolean value indicating if the track is motile or not, based on the minimum mean speed set by the user.
- [x] PERCENT_MOTILITY: the percentage of motile tracks in the image, calculated as the number of motile tracks divided by the total number of tracks times 100.

The file `spots_nameofinpoutimages.csv` contains the following columns:

- [x] LABEL: the label of the spot.
- [x] ID: the unique identifier of the spot.
- [x] TRACK_ID: the ID of the track the spot belongs to.
- [x] QUALITY: the quality of the spot detection, as calculated by Trackmate.
- [x] POSITION_X: the x-coordinate of the spot in pixels.
- [x] POSITION_Y: the y-coordinate of the spot in pixels.
- [x] POSITION_Z: the z-coordinate of the spot in pixels.
- [x] POSITION_T: the time point of the spot in seconds.
- [x] FRAME: the frame number of the spot.
- [x] RADIUS: the radius of the spot in pixels.
- [x] VISIBILITY: the visibility of the spot.
- [x] MANUAL_SPOT_COLOR: the manually assigned color of the spot.
- [x] MEAN_INTENSITY_CH1: the mean intensity of the spot in channel 1.
- [x] MEDIAN_INTENSITY_CH1: the median intensity of the spot in channel 1.
- [x] MIN_INTENSITY_CH1: the minimum intensity of the spot in channel 1.
- [x] MAX_INTENSITY_CH1: the maximum intensity of the spot in channel 1.
- [x] TOTAL_INTENSITY_CH1: the total intensity of the spot in channel 1.
- [x] STD_INTENSITY_CH1: the standard deviation of the intensity of the spot in channel 1.
- [x] CONTRAST_CH1: the contrast of the spot in channel 1.
- [x] SNR_CH1: the signal-to-noise ratio of the spot in channel 1.

For more information on the feature columns of the result file, you can consult the Trackmate documentation.

	A	B	C	D	E	F	G	H	I	J	K	L
1	LABEL	TRACK_ID	NUMBER_SP	NUMBER_GA	TRACK_DUR	TRACK_DISP	TRACK_MEA	TOTAL_DIST	MEAN_STRA	LINEARITY_C	MOTILE	PERCENT_M
2	Label	Track ID	Number of s	Number of g	Track durati	Track displa	Track mean	Total distanc	Mean straig	Linearity of forward progression		
3	Label	ID	N spots	N gaps	Duration	Track disp.	Mean sp.	Total dist.	Mn. v. line	Fwd. progr.		
4					(frames)	(pixels)	(pixels/fram	(pixels)	(pixels/frames)			
5	Track_0	0	100	0	1.980512	0.2160467	3.4227932	6.7788829	0.1090863	0.0318705	0	71.495327
6	Track_1	1	100	0	1.980512	0.1635081	8.0037727	15.851568	0.0825585	0.0103149	1	71.495327
7	Track_2	2	100	0	1.980512	0.3866606	4.7859099	9.4785519	0.1952327	0.0407932	0	71.495327
8	Track_3	3	95	5	1.980512	0.3643192	17.653458	34.091792	0.183952	0.0104202	1	71.495327
9	Track_4	4	100	0	1.980512	0.1690206	7.0412942	13.945367	0.0853419	0.0121202	1	71.495327
10	Track_5	5	100	0	1.980512	0.1730918	9.4555315	18.726793	0.0873975	0.009243	1	71.495327
11	Track_6	6	100	0	1.980512	0.2304839	5.9577487	11.799393	0.1163759	0.0195335	1	71.495327
12	Track_7	7	100	0	1.980512	0.6435074	10.288594	20.376683	0.3249197	0.0315806	1	71.495327
13	Track_8	8	100	0	1.980512	0.3201305	4.2587406	8.4344868	0.1616403	0.0379549	0	71.495327
14	Track_9	9	100	0	1.980512	0.3016632	5.1458251	10.191368	0.1523158	0.0295999	1	71.495327
15	Track_10	10	62	6	1.3603517	0.3172628	12.485476	15.681531	0.2332212	0.0186794	1	71.495327
16	Track_11	11	100	0	1.980512	0.4905724	14.816578	29.344409	0.2476998	0.0167177	1	71.495327
17	Track_12	12	87	7	1.9204965	0.1390553	13.448039	24.498225	0.0724059	0.0053841	1	71.495327

Hardware requirements

The plugin is designed to work with any time-lapse images of small round dots. Due to the size of the images, a computer with at least 16GB of RAM is recommended, more is preferred.